

Mean, median and mode; deviation and standard deviation

One number for the centre, one for the spread — and knowing which pair to use.

LESSON 68–70

3 × 40 MIN

ALGEBRA 11, §4.4

IGCSE E20.X

LESSON CLOCK

20'

26'

28'

28'

30'

8'

Three averages	20 min
Grouped data	26 min
Measures of spread	28 min
Standard deviation	28 min
Practice	30 min
Homework	8 min

LEARNING OBJECTIVES

- Compute the mean, median and mode of raw and of grouped data.
- Choose the appropriate average for a given distribution.
- Compute the range, the interquartile range and the standard deviation.
- Interpret the standard deviation as a typical distance from the mean.

TEXTBOOK

National	pp. 321–338
Cambridge	Core & Extended, pp. 397–412
Workbook	IGCSE Exercise 20.2

01

Explanation

Three averages, three uses

AVERAGE	DEFINITION	BEST WHEN	WEAKNESS
mean	$\frac{\sum x}{n}$	data is roughly symmetric	dragged by outliers
median	the middle value	data is skewed or has outliers	ignores most of the data
mode	the commonest value	data is categorical	may not exist or be unique

THE STANDARD ILLUSTRATION

Nine workers earn 4 million so'm and the director earns 60. The mean is 9.6 million — a figure nobody earns. The median is 4. For a question about a typical wage, the median is the honest answer.

Grouped data

The exact values are gone, so each class is represented by its **midpoint**:

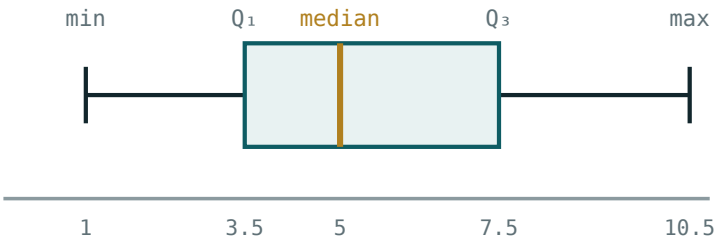
$$\text{estimated mean} = \frac{\sum fx}{\sum f} \text{ with } x \text{ the class midpoint}$$

CLASS	<i>F</i>	MIDPOINT <i>x</i>	<i>FX</i>
0–10	4	5	20
10–20	9	15	135
20–30	12	25	300
30–40	5	35	175
total	30		630

Estimated mean $\frac{630}{30} = 21$. The **modal class** is 20–30; the median lies in the class containing the 15th value, also 20–30.

Measures of spread

MEASURE	DEFINITION	RESISTANT TO OUTLIERS?
range	$max - min$	no — it is defined by them
interquartile range	$Q_3 - Q_1$	yes
standard deviation	see below	no



the box holds the middle half of the data

The five-figure summary. The box holds the middle half; its width is the interquartile range.

A **box plot** shows all five numbers at once, and makes two distributions comparable at a glance. A value more than $1.5 \times IQR$ beyond a quartile is conventionally called an **outlier** and marked separately.

Standard deviation

$$\sigma^2 = \frac{\sum(x - \bar{x})^2}{n} = \frac{\sum x^2}{n} - \bar{x}^2 \quad \sigma = \sqrt{\sigma^2}$$

The **variance** σ^2 is the mean of the squared deviations; the **standard deviation** σ is its square root, back in the original units.

WHY SQUARE, AND WHY THEN TAKE THE ROOT

The plain deviations always sum to zero, so they are useless. Squaring removes the signs; the square root brings the answer back to the units of the data, so that “ $\sigma = 4$ cm” is meaningful.

Example. 2, 4, 4, 4, 5, 5, 7, 9: $\bar{x} = 5$, $\sum x^2 = 232$, $\sigma^2 = \frac{232}{8} - 25 = 4$, so $\sigma = 2$.

For grouped data the same formula uses f and midpoints: $\sigma^2 = \frac{\sum fx^2}{\sum f} - \bar{x}^2$.

THE SECOND FORM IS THE ONE TO USE

$\frac{\sum x^2}{n} - \bar{x}^2$ needs one pass through the data; the definition needs two, because the mean must be known first. Both give the same number.

02 Worked examples

EXAMPLE 1 Find the mean, median, mode and standard deviation of 2, 4, 4, 4, 5, 5, 7, 9.

$$① \quad \bar{x} = \frac{40}{8} = 5$$

$$② \quad \text{Median } \frac{4+5}{2} = 4.5; \text{ mode } 4.$$

$$③ \quad \Sigma x^2 = 4+16+16+16+25+25+49+81 = 232$$

$$④ \quad \sigma^2 = 29 - 25 = 4 \Rightarrow \sigma = 2$$

ANSWER

$\bar{x} = 5$, median 4.5, mode 4, $\sigma = 2$

EXAMPLE 2 Estimate the mean of the grouped table with midpoints 5, 15, 25, 35 and frequencies 4, 9, 12, 5.

$$① \quad \Sigma f = 30$$

$$② \quad \Sigma fx = 20 + 135 + 300 + 175 = 630$$

$$③ \quad \bar{x} = 21$$

ANSWER

21 — an estimate

EXAMPLE 3 Nine workers earn 4 million and the director 60. Which average should a union quote?

① Mean $\frac{96}{10} = 9.6$.

Nobody earns that.

② Median 4.

Nine of ten earn exactly that.

③ The distribution has an extreme outlier.

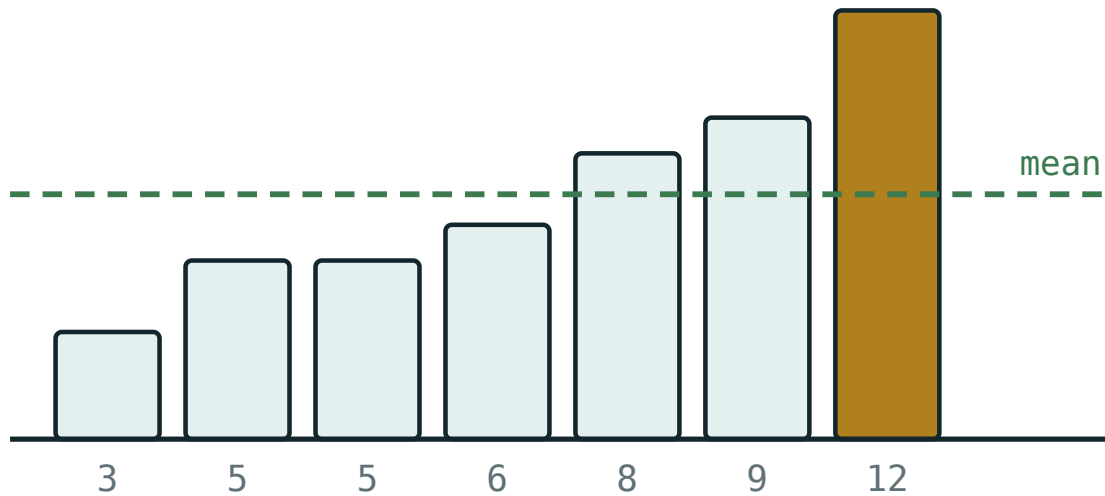
ANSWER

The median — the mean is distorted by one value

03 Interactive model

Compute the mean of the class's heights, then add one impossible value and recompute.

Centre and spread



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mean	O'rta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Modal class	Modal sinf	Модальный класс
Range	Kenglik	Размах
Interquartile range	Kvartillaro kenglik	Межквартильный размах
Deviation	Chetlanish	Отклонение
Variance	Dispersiya	Дисперсия
Standard deviation	Standart chetlanish	Стандартное отклонение
Outlier	Chetlashgan qiymat	Выброс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which average is dragged by an outlier?

the mean

the median

the mode

none

For skewed data the best average is:

the mean

the median

the mode

the range

σ^2 equals:

$$\frac{\sum x}{n}$$

$$\frac{\sum x^2}{n} - \bar{x}^2$$

$$\frac{\sum (x - \bar{x})}{n}$$

$$\sqrt{\sum x^2}$$

Why are deviations squared?

tradition

they otherwise sum to zero

to make them bigger

no reason

The interquartile range is:

$max - min$

$Q_3 - Q_1$

2σ

the median

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Mean of 3, 5, 7

ANSWER 5

2. Median of 3, 5, 7, 9

ANSWER 6

3. Mode of 2, 3, 3, 5

ANSWER 3

4. Range of 4, 9, 12

ANSWER 8

5. Mean of 2, 4, 4, 4, 5, 5, 7, 9

ANSWER 5

6. Median of that list

ANSWER 4.5

7. Mode of that list

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. σ of 2, 4, 4, 4, 5, 5, 7, 9

ANSWER 2

2. Estimated mean from midpoints 5,15,25,35, frequencies 4,9,12,5

ANSWER 21

3. The modal class there

ANSWER 20–30

4. σ of 1, 2, 3, 4, 5

ANSWER $\sqrt{2} \approx 1.41$

5. Nine at 4 and one at 60: the mean

ANSWER 9.6

6. Same: the median

ANSWER 4

7. $Q_3 - Q_1$ for 2,4,5,7,9,11,15

ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. Grouped: midpoints 5,15,25,35, $f = 4,9,12,5$. Find σ

ANSWER ≈ 8.6

2. A set of 10 has mean 12 and $\Sigma x^2 = 1560$. Find σ

ANSWER $\sqrt{12} \approx 3.46$

3. Adding 5 to every value changes the mean and σ how?

ANSWER Mean +5, σ unchanged

4. Multiplying every value by 3 changes them how?

ANSWER Both are tripled

5. A set has $\bar{x} = 20$, $\sigma = 4$. What are they after $y = 2x - 3$?

ANSWER 37 and 8

6. Is 25 an outlier for $Q_1 = 8$, $Q_3 = 14$?

ANSWER yes — beyond $14 + 9 = 23$

7. Explain why the median ignores most of the data but is still useful

ANSWER Its resistance is exactly that indifference

07 Homework

Homework — 6 tasks

Say which average you would quote, and why, on every real-context question.

- Find the mean, median, mode, range and standard deviation of 3, 5, 5, 6, 8, 8, 8, 9, 12, 16.
- Estimate the mean and standard deviation of grouped data with midpoints 2, 6, 10, 14 and frequencies 5, 12, 9, 4.
- Draw a box plot for 4, 7, 9, 11, 12, 15, 18, 22, 30 and identify any outlier.
- A firm has nine staff on 5 million and a director on 80. Compute both averages and say which is honest.

5. A set has $\bar{x} = 30$ and $\sigma = 6$. Find both after the transformation $y = 3x + 10$.
6. Explain in three sentences why the deviations are squared before averaging.